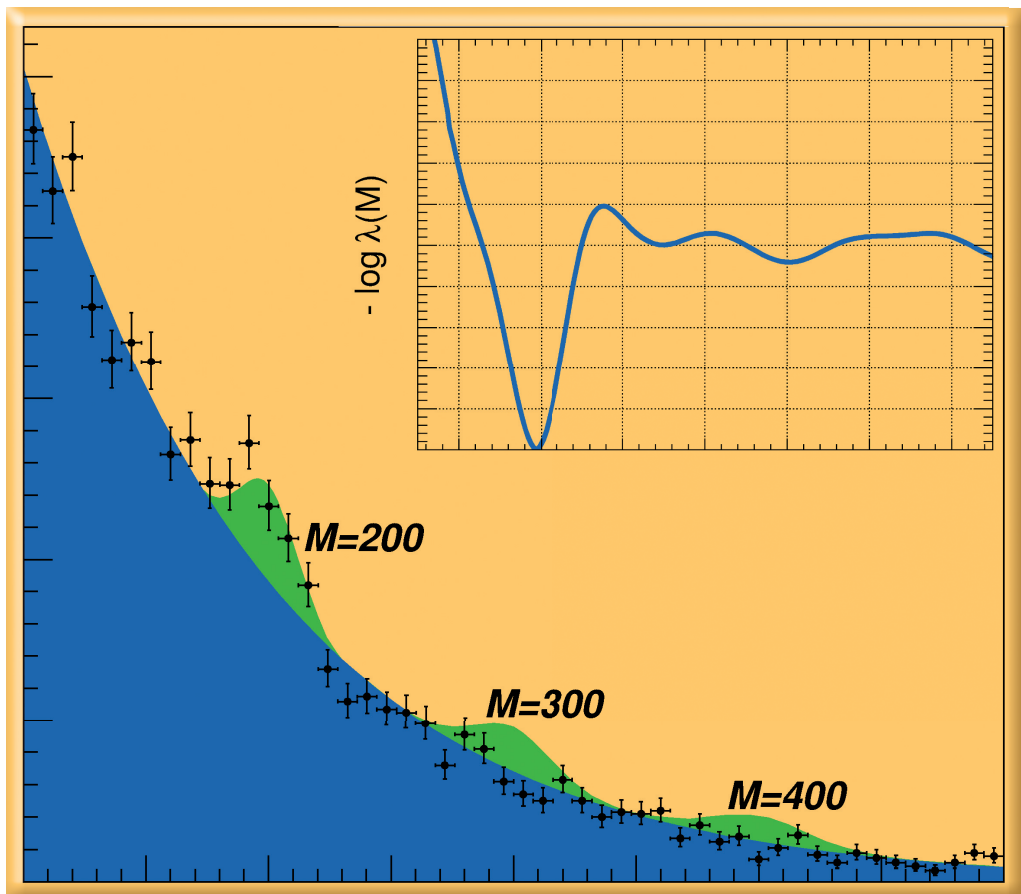


Edited by O. Behnke, K. Kröninger,
G. Schott, and T. Schörner-Sadenius

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Data Analysis in High Energy Physics

A Practical Guide to Statistical Methods



Edited by

*Olaf Behnke, Kevin Kröninger,
Grégory Schott, and
Thomas Schörner-Sadenius*

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*Olaf Behnke, Kevin Kröninger, Grégory Schott, and
Thomas Schörner-Sadenius*



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The Cover Picture

represents a hypothetical invariant-mass distribution. The markers with error bars represent the experimental data, the blue area the estimated background and the green regions possible signals for $M = 200$, $M = 300$ and $M = 400$ (in arbitrary units). The inset shows the negative logarithm of the likelihood function used to identify a resonance in the mass spectrum.

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Preface

Statistical inference plays a crucial role in the exact sciences. In fact, many results can only be obtained with the help of sophisticated statistical methods. In our field of experimental particle physics, statistical reasoning enters into basically every step of our data analysis work.

Recent years have seen the development of many new statistical techniques and of complex software packages implementing these. Consequently, the requirements on the statistics knowledge for scientists in high energy physics have increased dramatically, as have the needs for education and documentation in this field. This book aims at contributing to this purpose. It targets a broad readership at all career levels, from students to senior researchers, and is intended to provide comprehensive and practical advice for the various statistical analysis tasks typically encountered in high energy physics. To achieve this, the book is split into 12 chapters, all written by a different expert author or team of two authors and focusing on a well-defined topic:

- *Fundamental Concepts* introduces the basics of statistical data analyses, such as probability density functions and their properties, theoretical distributions (Gaussian, Poisson and many others) and concepts of probability (frequentist and Bayesian reasoning).

The next chapters elucidate the basic tools used to infer results from data:

- *Parameter Estimation* illustrates how to determine the best parameter values of a model from fitting data, for example how to estimate the strength of a signal.
- *Hypothesis Testing* lays out the framework that can be used to decide on hypotheses such as ‘the data can be explained by fluctuations of the known background sources alone’ or ‘the model describes the data reasonably well’.
- *Interval Estimation* discusses how to determine confidence or credibility intervals for parameter values, for example upper limits on the strength of a signal.

The following chapters deal with more advanced tasks encountered frequently:

- *Classification* presents various methods to optimally discriminate different event classes, for example signal from background, using multivariate data input. These methods can be very useful to enhance the sensitivity of a measurement, for example to find and measure a signal in the data that is otherwise drowned in background.
- *Unfolding* describes strategies and methods for correcting data for the usually inevitable effects of detector bias, acceptance, and resolution, which in particular can be applied in measurements of differential distributions.
- *Constrained Fits* discusses how to exploit physical constraints, such as energy–momentum conservation, to improve measurements or to determine unknown parameters.

The determination of systematic uncertainties is a key task for any measurement that is often performed as the very last step of a data analysis. We feel that it is worthwhile to discuss this – often neglected – topic in two chapters:

- *How to Deal with Systematic Uncertainties* elucidates how to detect and avoid sources of systematic uncertainties and how to estimate their impact.
- *Theory Uncertainties* illuminates various aspects of theoretical uncertainties, in particular for the strong interaction.

The following three chapters complete the book:

- *Statistical Methods Commonly Used in High Energy Physics* introduces various practical analysis tools and methods, such as the template and matrix methods for the estimation of sample compositions, or the determination of biases of analysis procedures by means of ensemble tests.
- *Analysis Walk-Throughs* provides a synopsis of the book by going through two complete analysis examples – a search for a new particle and a measurement of the properties of this hypothetical new particle.
- *Applications in Astronomy* takes us on a journey to the field of astronomy and illustrates, with several examples, the sophisticated data analysis techniques used in this research area.

In all chapters, care has been taken to be as practical and concrete as the material allows – for this purpose many specifically designed examples have been inserted into the text body of the chapters. A further deepening of the understanding of the book material can be achieved with the dedicated exercises at the end of all chapters. Hints and solutions to the exercises, together with some necessary software, are available from a webpage provided by the publisher. Here, we will also collect feedback, corrections and other information related to this volume; please check www.wiley.com for the details.

Many people have contributed to this book, and we would like to thank all of them. First of all, we thank the authors of the individual chapters for the high-quality material they provided.

Besides the authors, a number of people are needed to successfully conclude a book project like this one: numerous colleagues contributed by means of discussion, by providing expert advice and answers to our questions. We cannot name them all.

Katarina Brock spent many hours editing and polishing all the figures and providing a unified layout for them. Konrad Kieling from Wiley provided valuable support in typesetting the book. Vera Palmer and Ulrike Werner from Wiley provided constant support in all questions related to this book. We thank Tatsuya Nakada for his permission to use his exercise material.

Our last and very heartfelt thanks goes to our friends, partners and families who endured, over a considerable period, the very time- and also nerve-consuming genesis of this book. Without their support and tolerance this book would not exist today.

All comments, criticisms and questions you might have on the book are welcome – please send them to the authors via email:

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1

Fundamental Concepts*Roger Barlow***1.1****Introduction**

Particle physics is all about random behaviour. When two particles collide, or even when a single particle decays, we can't predict with certainty what will happen, we can only give probabilities of the various different outcomes. Although we measure the lifetimes of unstable particles and quote them to high precision – for the τ lepton, for example, it is 0.290 ± 0.001 ps – we cannot say exactly when a particular τ will decay: it may well be shorter or longer. Although we know the probabilities (called, in this context, branching ratios) for the different decay channels, we can't predict how any particular τ will decay – to an electron, or a muon, or various hadrons.

Then, when particles travel through a detector system they excite electrons in random ways, in the gas molecules of a drift chamber or the valence band of semi-conducting silicon, and these electrons will be collected and amplified in further random processes. Photons and phototubes are random at the most basic quantum level. The experiments with which we study the properties of the basic particles are random through and through, and a thorough knowledge of that fundamental randomness is essential for machine builders, for analysts, and for the understanding of the results they give.

It was not always like this. Classical physics was deterministic and predictable. Laplace could suggest a hypothetical demon who, aware of all the coordinates and velocities of all the particles in the Universe, could then predict all future events. But in today's physics the demon is handicapped not only by the uncertainties of quantum mechanics – the impossibility of knowing both coordinates and velocities – but also by the greater understanding we now have of chaotic systems. For predicting the flight of cannonballs or the trajectories of comets it was assumed, as a matter of common sense, that although our imperfect information about the initial conditions gave rise to increasing inaccuracy in the predicted motion, better information would give rise to more accurate predictions, and that this process could continue without limit, getting as close as one needed (and could afford) to

perfect prediction. We now know that this is not true even for some quite simple systems, such as the compound pendulum.

That is only one of the two ways that probability comes into our experiments. When a muon passes through a detector it may, with some probability, produce a signal in a drift chamber: the corresponding calculation is a *prediction*. Conversely a drift chamber signal may, with some probability, have been produced by a muon, or by some other particle, or just by random noise. To interpret such a signal is a process called *inference*. Prediction works forwards in time and inference works backwards. We use the same mathematical tool – *probability* – to cover both processes, and this causes occasional confusion. But the statistical processes of inference are, though less visibly dramatic, of vital concern for the analysis of experiments. Which is what this book is about.

1.2

Probability Density Functions

The outcomes of random processes may be described by a variable (or variables) which can be *discrete* or *continuous*, and a discrete variable can be *quantitative* or *qualitative*. For example, when a τ lepton decays it can produce a muon, an electron, or hadrons: that's a qualitative difference. It may produce one, three or five charged particles: that's quantitative and discrete. The visible energy (i.e. not counting neutrinos) may be between 0 and 1777 MeV: that's quantitative and continuous.

The probability prediction for a variable x is given by a function: we can call it $f(x)$. If x is discrete then $f(x)$ is itself a probability. If x is continuous then $f(x)$ has the dimensions of the inverse of x : it is $\int f(x)dx$ that is the dimensionless probability, and $f(x)$ is called a *probability density function* or *pdf*.¹⁾ There are clearly an infinite number of different pdfs and it is often convenient to summarise the properties of a particular pdf in a few numbers.

1.2.1

Expectation Values

If the variable x is quantitative then for any function $g(x)$ one can form the average

$$E[g] = \int g(x) f(x) dx \quad \text{or, as appropriate,} \quad E[g] = \sum g(x) f(x), \quad (1.1)$$

where the integral (for continuous x) or the sum (for discrete x) covers the whole range of possible values. This is called the *expectation value*. It is also sometimes written $\langle g \rangle$, as in quantum mechanics. It gives the mean, or average, value of g , which is not necessarily the most likely one – particularly if x is discrete.

1) The parton density functions of QCD, PDFs, share the abbreviation and are indeed pdfs in both senses.

1.2.2

Moments

For any pdf $f(x)$, the integer powers of x have expectation values. These are called the (algebraic) *moments* and are defined as

$$\alpha_n = E[x^n]. \quad (1.2)$$

The first moment, α_1 , is called the *mean* or, more properly, *arithmetic mean* of the distribution; it is usually called μ and often written $\bar{x}$. It acts as a key measure of *location*, in cases where the variable x is distributed with some known shape about a particular point.

Conversely there are cases where the shape is what matters, and the absolute location of the distribution is of little interest. For these it is useful to use the *central moments*

$$m_n = E[(x - \mu)^n]. \quad (1.3)$$

1.2.2.1 **Variance**

The second central moment is also known as the *variance*, and its square root as the *standard deviation*:

$$V[x] = \sigma^2 = m_2 = E[(x - \mu)^2]. \quad (1.4)$$

The variance is a measure of the width of a distribution. It is often easier to deal with algebraically whereas the standard deviation σ has the same dimensions as the variable x ; which to use is a matter of personal choice. Broadly speaking, statisticians tend to use the variance whereas physicists tend to use the standard deviation.

1.2.2.2 **Skew and Kurtosis**

The third and fourth central moments are used to build shape-describing quantities known as *skew* and *kurtosis* (or *curtosis*):

$$\gamma_1 = \frac{m_3}{\sigma^3}, \quad (1.5)$$

$$\gamma_2 = \frac{m_4}{\sigma^4} - 3. \quad (1.6)$$

Division by the appropriate power of σ makes these quantities dimensionless and thus independent of the scale of the distribution, as well as of its location. Any symmetric distribution has zero skew: distributions with positive skew have a tail towards higher values, and conversely negative skew distributions have a tail towards lower values. The Poisson distribution has a positive skew, the energy recorded by a calorimeter has a negative skew. A Gaussian has a kurtosis of zero – by definition, that's why there is a '3' in the formula. Distributions with positive

kurtosis (which are called *leptokurtic*) have a wider tail than the equivalent Gaussian, more centralised or *platykurtic* distributions have negative kurtosis. The Breit–Wigner distribution is leptokurtic, as is Student’s t . The uniform distribution is platykurtic.

1.2.2.3 Covariance and Correlation

Suppose you have a pdf $f(x, y)$ which is a function of two random variables, x and y . You can not only form moments for both x and y , but also for combinations, particularly the *covariance*

$$\text{cov}[x, y] = E[xy] - E[x]E[y]. \quad (1.7)$$

If the joint pdf is factorisable: $f(x, y) = f_x(x) \cdot f_y(y)$, then x and y are independent, and the covariance is zero (although the converse is not necessarily true: a zero covariance is a necessary but not a sufficient condition for two variables to be independent).

A dimensionless version of the covariance is the *correlation* ρ :

$$\rho = \frac{\text{cov}[x, y]}{\sigma_x \sigma_y}. \quad (1.8)$$

The magnitude of the correlation lies between 0 (uncorrelated) and 1 (completely correlated). The sign can be positive or negative: amongst a sample of students there will probably be a positive correlation between height and weight, and a negative correlation between academic performance and alcohol consumption.

If there are several (i.e. more than two) variables, $x_1, x_2, \dots, x_N$, one can form the *covariance* and *correlation matrices*:

$$V_{ij} = \text{cov}[x_i, x_j] = E[x_i x_j] - E[x_i]E[x_j], \quad (1.9)$$

$$\rho_{ij} = \frac{V_{ij}}{\sigma_i \sigma_j}, \quad (1.10)$$

and V_{ii} is just σ_i^2 .

1.2.2.4 Marginalisation and Projection

Mathematically, any pdf $f(x, y)$ is a function of two variables x and y . They can be similar in nature, for example the energies of the two electrons produced by a converting high energy photon, or they can be different, for example the position and direction of particles undergoing scattering in material.

Often we are really interested in one parameter (say x) while the other (say y) is just a *nuisance parameter*. We want to reject the extra information shown in the two-dimensional function (or scatter plot). This can be done in two ways: the *projection* of x , $f(x)|_y$ is obtained by choosing a particular value of y , the *marginal distribution* $f(x) = \int f(x, y)dy$ is found by integrating over y .

Projections can be useful for illustration, otherwise to be meaningful you have to have a good reason for choosing that specific value of y . Marginalisation requires that the distribution in y , like that of x , is properly normalised.

1.2.2.5 Other Properties

There are many other properties that can be quoted, depending on the point we want to bring out, and on the established usage of the field.

The mean is not always the most helpful measure of location. The *mode* is the value of x at which the pdf $f(x)$ is maximum, and if you want a typical value to quote it serves well. The *median* is the midway point, in the sense that half the data lie above and half below. It is useful in describing very skewed distributions (particularly financial income) in which fluctuations in a small tail would give a big change in the mean.

We can also specify dispersion in ways that are particularly useful for non-Gaussian distributions by using *quantiles*: the upper and lower *quartiles* give the values above which, and below which, 25% of the data lie. *Deciles* and *percentiles* are also used.

1.2.3

Associated Functions

The *cumulative distribution function*

$$F(a) = \int_{-\infty}^a f(x)dx \quad \text{or, as appropriate,} \quad F(a) = \sum f(x_i)\Theta(a - x_i), \quad (1.11)$$

where Θ is the Heaviside or step function ($\Theta(x) = 1$ for $x \geq 0$ and 0 otherwise), giving the probability that a variable will take a value up to a , is occasionally useful.

The *characteristic function*

$$\phi(u) = E[e^{iux}] = \int e^{iux} f(x)dx, \quad (1.12)$$

which is just (up to factors of 2π) the Fourier transform of the pdf, is also met with sometimes as it has useful properties.

1.3

Theoretical Distributions

A pdf is a mathematical function. It involves a variable (or variables) describing the random quantity concerned. This may be a discrete integer or a continuous real number. It also involves one or more parameters. In what follows we will denote a random variable by x for a real number and r for an integer. Parameters generally have their traditional symbols for particular pdfs: where we refer to a generic parameter we will call it θ . It is often helpful to write a function as $f(x; \theta)$ or $f(x|\theta)$, separating this way more clearly the random variable(s) from the adjustable parameter(s). The semicolon is preferred by some, the line has the advan-

tage that it matches the notation used for conditional probabilities, described in Section 1.4.4.1.

There are many pdfs in use to model the results of random processes. Some are based on physical motivations, some on mathematics, and some are just empirical forms that happen to work well in particular cases.

The overwhelmingly most useful form is the *Gaussian* or *normal* distribution. The *Poisson* distribution is also encountered very often, and the *binomial* distribution is not uncommon. So we describe these in some detail, and then some other distributions rather more briefly.

1.3.1

The Gaussian Distribution

The Gaussian, or normal, distribution for a continuous random variable x is given by

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \quad (1.13)$$

It has two parameters; the function is manifestly symmetrical about the location parameter μ , which is the mean (and mode, and median) of the distribution. The scale parameter σ is also the standard deviation of the distribution. So there is, in a sense, only one Gaussian, the *unit Gaussian* or *standard normal distribution* $f(x; 0, 1)$ shown in Figure 1.1. Any other Gaussian can be obtained from this by scaling by a factor σ and translating by an amount μ . The Gaussian distribution is sometimes denoted $\mathcal{N}(x; \mu, \sigma)$.

The Gaussian is ubiquitous (hence the name ‘normal’) because of the *central limit theorem*, which states that if any distribution is convoluted with itself a large number of times, the resulting distribution tends to a Gaussian form. For a proof, see for example Appendix 2 in [1].

Gaussian random numbers are much used in simulation, and a suitable random number generator is available on most systems. If it is not, then you can generate a unit Gaussian by taking two uniformly generated random numbers u_1, u_2 , set

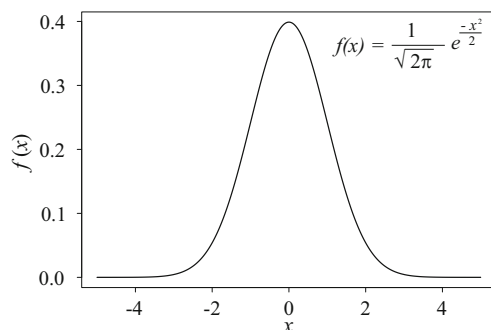


Figure 1.1 The unit Gaussian or standard normal distribution.

$\theta = 2\pi u_1$, $r = \sqrt{-2 \ln u_2}$, and then $r \cos \theta$ and $r \sin \theta$ are independent samples from a unit Gaussian.

The product of two independent Gaussians gives a two-dimensional function

$$f(x, y; \mu_x, \mu_y, \sigma_x, \sigma_y) = \frac{1}{2\pi\sigma_x\sigma_y} e^{-\frac{1}{2}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]}, \quad (1.14)$$

but the most general quadratic form in the exponent must include the cross term and can be written as

$$f(x, y; \mu_x, \mu_y, \sigma_x, \sigma_y, \rho) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{(1-\rho^2)}} e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 - \frac{2\rho(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]}, \quad (1.15)$$

where the parameter ρ is the correlation between x and y . For N variables, for which we will use the vector $\mathbf{x}$, the full form of the multivariate Gaussian can be compactly written using matrix notation:

$$f(\mathbf{x}; \boldsymbol{\mu}, \mathbf{V}) = \frac{1}{(2\pi)^{N/2}|\mathbf{V}|} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \mathbf{V}^{-1}(\mathbf{x}-\boldsymbol{\mu})}. \quad (1.16)$$

Here, $\mathbf{V}$ is the covariance matrix described in Section 1.2.2.3.

The *error function* and the *complementary error function* are basically closely related to the cumulative Gaussian

$$\text{erf}(y) = \frac{2}{\sqrt{\pi}} \int_0^y e^{-x^2} dx, \quad (1.17)$$

$$\text{erfc}(y) = \frac{2}{\sqrt{\pi}} \int_y^\infty e^{-x^2} dx. \quad (1.18)$$

Their main use is in calculating Gaussian *p-values* (see Section 1.3.4.6). The probability that a Gaussian random variable will lie within one standard deviation, or ‘1 σ ’, of the mean is 68% obtained by calculating $\text{erf}(y = 1)$. Conversely, the chance that a variable drawn from a Gaussian random process will lie outside 1 σ is 32%. Given such a process – say a mean of 10.2 and a standard deviation of 3.1 – then if you confront a particular measurement – say 13.3 – it is quite plausible that it was produced by the process. One says that its *p-value*, the probability that the process would produce a measurement this far, or further, from the ideal mean, is 32%. Conversely, if the number were 25.7 rather than 13.3, that would be 5 σ rather than 1 σ , for which the *p-value* is only $5.7 \cdot 10^{-7}$. In discussion of discoveries (or otherwise) of new particles and new effects this language is turned round, and a discovery with a *p-value* of $5.7 \cdot 10^{-7}$ is referred to as a ‘5 σ result’²⁾. A translation is given in Table 1.1 – although for practical purposes it is easier to use functions such as `pnorm` and `qnorm` in the programming language `R` [2], or `TMath::Prob` in `ROOT` [3].

2) Note that there is a subtle difference between a one-sided and two-sided *p-value*. Details will be discussed in Chapter 3

Table 1.1 Two-sided Gaussian p -values for 1σ to 5σ deviations.

Deviation	p -value (%)
1σ	31.7
2σ	4.56
3σ	0.270
4σ	0.006 33
5σ	0.000 057 3

1.3.2

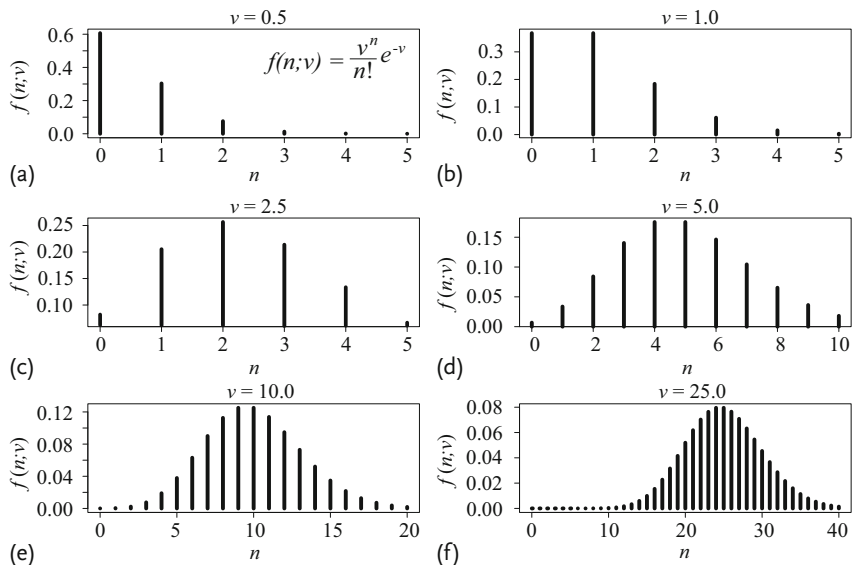
The Poisson Distribution

The Poisson distribution

$$f(n; \nu) = \frac{\nu^n}{n!} e^{-\nu} \quad (1.19)$$

describes the probability of n events occurring when the mean expected number is ν ; n is discrete and ν is continuous. Typical examples are the number of clicks produced by a Geiger counter in an interval of time, or, famously, the number of Prussian cavalymen killed by horse-kicks [4]. Some examples are shown in Figure 1.2.

The Poisson distribution has a mean of ν and a standard deviation $\sigma = \sqrt{\nu}$. This property – that the standard deviation is the square root of the mean – is a key

**Figure 1.2** Poisson distributions for (a) $\nu = 0.5$, (b) $\nu = 1.0$, (c) $\nu = 2.5$, (d) $\nu = 5.0$, (e) $\nu = 10.0$, (f) $\nu = 25.0$.